

Michael Williams de la Bastida

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PUBLICATIONS

J=JOURNAL, P=IN PREPARATION, R=IN REVIEW

- [R.2] Williams de la Bastida, M, Coveney, P.V., **Parallel and Distributed Fermionic Simulation via Dynamic Encoding**. arXiv:2609.31502
- [R.1] Orlov, A., et. al. **Quantum Computing and GPU-Accelerated Reaction-Barrier Calculations for aGPCR Autoproteolysis**.
- [J.3] Williams de la Bastida, M., Bickley, TM., Coveney, PV., (2025). **Optimised fermion-qubit encodings for quantum simulation with reduced circuit depth**. *IOP Quantum Science and Technology*, DOI: 10.1088/2058-9565/ae9b40
- [J.2] Bickley, TM., et al. (2025). **Extending quantum computing through subspace, embedding and classical molecular dynamics techniques** . *Digital discovery*, Vol. 4, Issue 12, pp. 3427-3444. DOI: 10.1039/D5DD00225G
- [J.1] Ralli A., Williams de la Bastida, M., Coveney, PV., (2024). **Scalable approach to quantum simulation via projection-based embedding** . In *Physical Review A*, 109, 022418, DOI: 10.1103/PhysRevA.109.022418

SERVICE

- **Journal of Open Source Software** 1 Review in 2026
Reviewer - Quantum Computing, Computational Chemistry, Python, Rust, R

EDUCATION

- **University College London** In Progress (Completion 2026)
PhD Chemistry - Resource Reduction for Hybrid Quantum-HPC Simulation of Chemistry London, England
- **University College London** 2021
MRes Quantum Technologies - Embedded Quantum Algorithms for Chemical Structure London, England
 - Distinction
- **University of St Andrews** 2019
MPhys Theoretical Physics - Examining the Orthogonality Catastrophe with the Kernel Polynomial Method St Andrews, Scotland
 - First Class
- **Flight Training Europe** 2013
Airline Transport Pilot's License Jerez, Spain
 - 93% Average in 14 ATPL theory exams
 - 1st attempt pass Commercial Pilot's License and Instrument Rating
 - Jet Orientation Course with Multi-crew Co-ordination

EXPERIENCE

- **Microsoft** September 2019- September 2020
Automation Software Engineer London, England
 - Developed cloud automation tools for deployment and life-cycle management of telecoms virtual machines in Python and Rust.
 - Implemented testing and data-validation models for cloud infrastructure.
 - Provided installation engineers with CLI tools for headless debugging, resulting in reduced commissioning time.
- **ProEd Et Al** July 2023 - Present
Lecturer
 - **Introduction to Quantum Computing from Computer Science**: 6 Hour in-person course for university entrants.
 - **Physical Sciences and Programming** 3 Day course covering study and careers in physical and information sciences for students in late secondary education.
 - **Introduction to Computer Science**, 6 Hour online course covering classical information theory, physical computing and computational science for students in secondary education.

SKILLS

- **Research Skills**: Quantum algorithms, Simulation of chemistry, Resource reduction of quantum algorithms, Education and Dissemination
- **Programming Languages**: Python, Rust, R
- **Cloud Technologies**: Terraform, Ansible
- **DevOps & Version Control**: Git, Jujutsu, GitHub Actions